



Fiona Stanley Fremantle Hospitals Group

Cognitive screening of the older person

Reference #: FSFH-HW-POL-0028

Scope

Site	Service/Department/Unit	Disciplines
Fiona Stanley Hospital and Fremantle Hospital	Hospital Wide	Medical, Nursing, and Allied Health
Exclusions: Patients admitted with a new acquired brain injury requiring impaired cognition observations or requiring Post Traumatic Amnesia (PTA) testing.		

1. Introduction

Cognitive screening allows the detection and appropriate treatment of people with Cognitive Impairment (CI). CI (dementia and delirium) is common in hospitalised older people, who are at high risk of experiencing adverse events. This document outlines the screening and care required for all inpatients at Fiona Stanley and Fremantle Hospitals Group (FSFHG) over 65 years or Aboriginal people over 45 years presenting to the FSFHG Emergency Department (ED) and through the booked or direct admission process. When clinically appropriate, patients under the age of 65 (or 45 for Aboriginal people) demonstrating difficulties with memory and thinking are suitable for inclusion within this policy.

2. Terminology

Abbreviated Mental Test – 4 item (AMT4)	A practical screening tool for cognitive impairment for older patients within a busy clinical environment. It provides a brief assessment of cognitive function (Appendix 1).
Cognition	The set of higher cortical brain functions that includes memory, thinking, planning, problem solving, orientation, comprehension, calculation, learning capacity, language and judgement/intuition.
Cognitive Impairment (CI)	It is acknowledged that the term cognitive impairment is inclusive of a broad range of genetic, medical and acquired conditions and can affect people from any age.

Communication disorder	Impairment in the ability to receive, send, process, and comprehend concepts or verbal, nonverbal and graphic symbol systems. A communication disorder may be evident in the processes of hearing, language, and/or speech.
Dementia	A syndrome due to disease of the brain, usually of a chronic or progressive nature, in which there is a disturbance of multiple higher cortical function, including memory, thinking, orientation, comprehension, calculation, learning capacity, language and judgement. Consciousness is not clouded.
Delirium	Characterised by a disturbance of consciousness, attention, cognition and perception that develops over a short period of time (usually hours to a few days).
Falls Risk Assessment and Management Plan plus Cognitive Impairment (FRAMP-CI)	The Falls Risk Assessment and Management Plan plus Cognitive Impairment form is an interdisciplinary screening, assessment, and intervention plan.
Four A's Test (4AT)	A brief screening tool designed for the detection of possible delirium as well as cognitive impairment. ¹ This tool has 4 separate domains which measure alertness, AMT 4, attention and acute change/fluctuating course. The tool has a maximum score of twelve with a higher score indicating increased impairment (Appendix 2). ² The 4AT has been embedded on page 4 of the FRAMP-CI.

3. Policy

- All at risk patients as defined by the inclusion criteria in this policy will have their cognition screened on presentation and direct or booked admission to the hospital using the Abbreviated Mental Test – 4 Item (AMT4) screening tool ([Appendix 1](#)). This can be located on the ED triage form, Fremantle Hospital's patient admission form and question 3 in the Falls Risk Assessment and Management Plan plus Cognitive Impairment (FRAMP-CI).
- When the patient is part of the ED Falls Pathway the Four A's Test (4AT) screening tool ([Appendix 2](#)) will also be completed in the ED. Staff will be alert to concerns raised by the patient or their carer, regarding any recent changes in behaviour and/or thinking, which could indicate a CI and any change they notice during their shift.
- Staff will document the impact that the CI has had on the patient's care needs.

4. Procedure

The AMT4 and the 4AT are to be used with patients across Fiona Stanley and Fremantle Hospitals.

- These tools can be used to detect cognitive impairment: they are not diagnostic assessments.
- Results are affected by the responders' educational and cultural background.
- The results can be biased by various communication disorders.

When a patient's AMT4 score is less than 4/4, a series of actions will follow. Patients will have the Cognitive Impairment Identifier (CII) placed at their bedside ([Appendix 3](#)) and all staff will implement the FSFHG communication and carer engagement strategies ([Appendix 4](#)). For patient safety, a CII can be placed in a patient's room if they have another known cognitive impairment that is not dementia or delirium, they do not have to be within the age range stipulated in the policy. The CII can be provided for any ONE of the following reasons:

- The patient has a known history of cognitive impairment (previous traumatic brain injury or stroke, dementia, intellectual disability).
- Acute confusion is present on admission or develops during admission.
- A carer reports memory /thinking difficulties.

The CII may be taken down if requested by a patient or carer, however, it is important to ensure that they are aware that it is being implemented to improve the care and safety of patients. Ensure that they have been provided with the CII information brochure. See [Appendix 5](#) for the Cognitive Screening, Delirium Identification and Management Flow Chart.

4.1. Admitted Patients

4.1.1. Recording and handover of cognitive screening scores:

- ED nursing and/or medical staff will record the AMT4 score in the ED triage form; identify the risks and action appropriate ED risk minimisation strategies.
- ED nursing staff to note the patients AMT4 scores in the falls section of the WA Health Enterprise Bed Management (EBM) application to ensure that handover to wards regarding the presence of CI is consistent.
- On arrival to the receiving ward, nursing staff will complete the FRAMP-CI and Patient Care Plan. If the documented AMT4 is less than 4/4 the admitting nurse will be required to complete the 4AT on page 4 of the FRAMP-CI.
- The clinical staff member who identifies the risk will provide patient, carer, and family with CII information brochure and secure the CII in the specified location for the ward.
- Clinical staff members who completed the AMT4 or 4AT will document results in the observation section of the iSoBAR handover for the admitting ward and all

subsequent inter-ward transfer summary documents as well as ensuring the CII is on the patients bed so patients can be identified as they move through the hospital. The CII is to be removed from a patient's room on discharge.

4.1.2. Ongoing Assessment

- BE ALERT TO DELIRIUM if the patient scores 4 or above on the 4AT section of the FRAMP-CI with no previous diagnosed CI, scores higher than previously documented in the same admission, or there are carer concerns.
 - The nurse will refer to the medical team for review and document the score in the patient's medical record.
 - Patients and carers will be engaged and provided with the [Delirium Brochure](#)
 - The patient will then be assessed for the cause of their acute change in cognitive status by their medical team with input from allied health and nursing.
- BE ALERT TO DEMENTIA if the patient scores 1, 2 or 3 on the 4AT section of the FRAMP-CI with no previously diagnosed CI it is possible that there is an underlying condition that should be addressed whilst in hospital.
- All staff will observe patients for signs of cognitive deficits interfering with their activities of daily living and notify the medical team and occupational therapists. Any concerns or referrals will then be documented in the patient's medical record.
- All staff will implement interventions to prevent delirium including those listed in the Cognitive State section of the FRAMP-CI and any other suitable person-centred interventions identified by the multidisciplinary team.
- The Occupational Therapist will review the patient's cognitive and functional status to assist with care planning during admission and to determine the patient's individual, occupational and environmental needs for discharge.
- Occupational therapists or medical staff will determine the clinical need for a detailed cognitive screen and complete the screen based on their presentation and cultural appropriateness (this includes the use of Kimberly Indigenous Cognitive Assessment (KICA) and the Rowland Universal Dementia Assessment Scale (RUDAS)):
 - Staff completing the cognitive screen must ensure that an interpreter has been booked through language services for patients from Culturally And Linguistically Diverse backgrounds.
 - For cognitive screening of Aboriginal people, staff must gain consent from the patient or family prior to contacting an Aboriginal Hospital Liaison Officer (AHLO) to be present. A booking should also be made for an interpreter through Language Services if the patient has a limited understanding of English.

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- The medical team will refer to the Geriatrics service via e-referral if the delirium is not resolving within 24 hours or if the patient is demonstrating behavioural and psychological symptoms of dementia that are impacting on care provision.
 - The multidisciplinary team will continue to be alert to delirium and observe patients for acute changes in consciousness, behaviour and thinking.
 - The multidisciplinary team will be alert to the increased risk of other hospital complications for the patient including the increased risk of functional decline, falls, pressure injuries, malnutrition and dehydration using local screening tools for these risks and providing interventions when required.
 - Staff will give information to families of people with a CI on how they can escalate concerns about changes in the patient's behaviour or cognitive status and document this interaction in the patient's medical record.
 - The multidisciplinary team will refer to Social Work as soon as possible if there are concerns for the patient's ability to manage at home on discharge.

4.1.3. Discharge Planning

- On discharge from ward areas, the medical team will highlight the presence of CI and episodes of delirium in the discharge summary within the Notifications and Clinical Summaries (NaCS) application for the GP to follow up.
- If the patient has not been seen by a Geriatrician during admission this should be suggested to the GP to consider a referral post discharge.
- Please see the [Discharge Summary Guide for Cognitive Impairment and Falls](#) for more information.

4.2 Patients Discharged from the Emergency Department

- Medical staff must ensure the CI is not due to new onset delirium before discharge
- All ED patients with an AMT4 score of less than 4/4 who are to be discharged, require documentation in their discharge summary (NACS) recommending referral to a memory clinic and CI follow up by the GP.
- Please see the [Discharge Summary Guide for Cognitive Impairment and Falls](#) for more information.

- If the patient is transferred to another hospital, their CI will be included in the observation section of the verbal and written iSoBAR handover with recommendations to re-screen cognition on arrival at the hospital.

5. Compliance/Performance Monitoring

The Comprehensive Care - Cognitive Impairment Committee is responsible for monitoring and reporting compliance to this document.

- Compliance of this policy will be monitored through the Weekly Engagement of Consumers in an Audit of National Standards (WECAN), an online audit tool that was rolled out across FSFHG in late 2021. The audit is designed to measure compliance against the National Safety and Quality Health Service Standards and reflect the clinical flow when assessing a patient.
- A WECAN audit report will be tabled at the Comprehensive Care – Cognitive Impairment Committee quarterly, with a quality improvement plan included if warranted. When required, this will be escalated to the Comprehensive Care Steering Committee, the Service Four Safety and Quality Committee and the FSFHG Clinical Governance Committee for evaluation.
- Other individual audits will be completed if the need is identified by the Comprehensive Care – Cognitive Impairment Committee.

6. Related Policy Documents

- FSFHG Management of Agitation in Older Adults with Dementia or Delirium
- FSFHG Falls Prevention and Management Policy

7. Related Standards

National Safety and Quality Health Service Standards (Second Edition)

- Clinical Governance Standard
- Partnering with Consumers Standard
- Preventing and Controlling Healthcare-Associated Infection Standard
- Medication Safety Standard
- Comprehensive Care Standard
 - [Action 5.29](#)
 - [Action 5.30](#)
- Communicating for Safety Standard
- Recognising and Responding to Acute Deterioration Standard
 - [Action 8.05](#)

Delirium Clinical Care Standard (2021)

8. Bibliography

1. Bellelli G, Morandi A, Davis D, Mazzola P, Turco R, Gentile S, Ryan T, Cash H, Guerini F, Torpilliesi T, Del Santo F, Trabucchi M, Annoni G, MacLulich A; Validation of the 4AT, a new instrument for rapid delirium screening: a study in 234 hospitalised older people. Age and Ageing [internet], 2014 July [cited 2018 Feb 10]; 43(4):496–502. Available from: <https://doi.org/10.1093/ageing/afu021>
2. Lees R, Corbet S, Johnston C, Moffitt E, Shaw G, Quinn TJ. Test accuracy of short screening tests for diagnosis of delirium or cognitive impairment in an acute stroke unit setting. Stroke. 2013; 29:1-6.

Other related documents:

- A Better Way to Care: Safe and high-quality care for patients with cognitive impairment (dementia and delirium) in hospital
- Clinical Practice Guidelines for Dementia in Australia

9. Authorisation

EXECUTIVE SPONSOR: Nurse Director, Service 4					
Version	Date Issued	Compiled/Revised By	Committee/Consumer Group Consulted	Endorsed By	Revision due
1	10/2014	Snr Project Officer, FSH Clinical Commissioning		Co-Director Service 1	03/2018
2	09/2018	Snr Project Officer, FSFHG CI Project	CGC	FSFHG Policy Committee	09/2020
3	05/2022	FSFHG Comprehensive Care - Cognitive Impairment Committee	SMHS Aboriginal Health Strategy	FSFHG Policy Committee	05/2025
3.1	03/2023	Policy Team	Review date smoothing	HEC PPC	06/2026

10. Appendices

10.1 Appendix 1: 4-item Abbreviated Mental Test (AMT4)

10.2 Appendix 2: Four A's Test (4AT)

10.3 Appendix 3: Cognitive Impairment Identifier (CII)

10.3 Appendix 4: FSFHG Communication and Carer Engagement Strategies

10.4 Appendix 5: Cognitive Screening, Delirium Identification and Management Flow Chart

10.1. Appendix 1:

Abbreviated Mental Test – 4 Item (AMT4)

4 Questions:

1. How old are you?
2. What is your date of birth?
3. What year is it?
4. Where are we?

Each correctly answered question scores 1 point.

A score of less than 4/4 indicates possible Cognitive Impairment.

The AMT4 is present in the ED triage form and also appears within Question 3 on the FRAMP Form.

10.2 Appendix 2

Four A's Test (4AT)

		Initial screen	Rescreen 1	Rescreen 2
DATE				
4AT / DELIRIUM OR COGNITIVE IMPAIRMENT RISK ASSESSMENT Complete only if delirium / cognitive risks have been identified				
1	ALERTNESS			
Patient markedly drowsy (e.g. difficult to rouse and/or obviously sleepy during assessment) or agitated / hyperactive . Observe patient if asleep, attempt to wake. Ask patient to state name and address to assist rating.				
Normal: fully alert, but not agitated		0	0	0
Mild sleepiness: for <10 seconds after waking, then normal		0	0	0
Clearly abnormal: significantly altered alertness		4	4	4
2	AMT4			
Age Date of Birth Current year Place (name of hospital or building) (you can transfer this score from page 1 if completed at the same time as this risk screen)				
AMT 4/4: no mistakes		0	0	0
AMT 3/4: 1 mistake		1	1	1
AMT 0-2/4: 2 or more mistakes / untestable		2	2	2
3	ATTENTION			
Ask the patient: "Please tell me the months of the year in backwards order, starting at December." To assist initial understanding, one prompt of "What is the month before December?" is permitted.				
Achieves 7 months or more correctly:		0	0	0
Starts but scores <7 months / refuses:		1	1	1
Untestable: cannot start because unwell, drowsy, inattentive		2	2	2
4	ACUTE CHANGE or FLUCTUATING COURSE			
Evidence of significant change or fluctuation in alertness, cognition, other mental function (eg hallucinations, paranoia) arising over the last 2 weeks and still evident in last 24 hours. <i>Fluctuation can occur without delirium in some cases of dementia, but marked fluctuation usually indicates delirium</i>				
No:		0	0	0
Yes:		4	4	4
4AT score:				
4AT / DELIRIUM OR COGNITIVE IMPAIRMENT RISK SCORE and ACTIONS				
		Initial actioned interventions		
4AT = 0	Delirium or severe cognitive impairment unlikely (delirium still possible if question 4 incomplete)			
Alert Medical Officer if 4AT score is ≥ 1 OR if 4AT score is HIGHER than admission score				
4AT = 1-3	Possible cognitive impairment Ensure delirium prevention strategies are in place and patient is referred to OT			
4AT ≥ 4	Possible delirium +/- cognitive impairment Ensure delirium prevention strategies are in place and patient is referred to OT			
Medical review required regardless of pre-existing CI. Medical staff documentation must acknowledge presence of cognitive impairment and/or delirium and have documented strategies to support the patient				
Please note this risk screen has limitations for Aboriginal Torres Strait Islanders and patients who speak English as a second language. Consider assistance of ALO or Interpreters to complete.				
© 2011-2014 MacLulich, Ryan, Cash Version 1.2 Information and download: www.the4AT.com				

10.3 Appendix 3: Cognitive Impairment Identifier (CII)



10.4 Appendix 4:

FSFHG Communication and Carer Engagement Strategies

- Introduce yourself
- Make sure you have eye contact at all times
- Remain calm and talk in a matter of fact way
- Involved carers
- Keep sentences short and simple
- Focus on one instruction at a time
- Give time for responses
- Repeat yourself... don't assume you have been understood
- Do not give too many choices

10.5 Appendix 5:

Cognitive Screening, Delirium Identification and Management Flow Chart

Cognitive Screening, Delirium Identification & Management Flow Chart

See also "Cognitive screening of the older adult" policy Appendix 5
(location: [FSFHG Policy page](#), link also on [Dementia and Delirium hub page](#))

